MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.2: One-parameter maps and changes

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Vector-valued functions of one variable

Basic definitions

A map $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ takes one scalar parameter and outputs a vector. It traces a curve.
Unit 2 lines are the special case $\mathbf{r}(t) = \mathbf{x}_0 + t\mathbf{d}$. Indeed, we already encountered parameterizations of curves when discussing parametric equations for straight lines, as in Parametric equation for a line.

Activity: Parametrizing two lines (U3-LO1)

The line L_1 passes through point $(3, -2, 5)$ and is parallel to the vector $\mathbf{v} = (2, -3, 4)$.
Find a parameterization of the line L_1.

The line passes through $(3, -2, 5)$ and has direction vector $(2, -3, 4)$, and so it parameterized by the function

$$\mathbf{r}(t) = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 + 2t \\ -2 - 3t \\ 5 + 4t \end{bmatrix}.$$

Activity: Parametrizing two lines (U3-LO1)

The line L_1 passes through point $(3, -2, 5)$ and is parallel to the vector $\mathbf{v} = (2, -3, 4)$.
Another line L_2 passes through the point $(1, 3, -1)$ and is perpendicular to L_1. If L_2 also lies in the plane $z = -1$, find a parameterization of L_2.

Since L_2 lies in the plane $z = -1$, the z-coordinates of all the points of L_2 are the same, so any direction vector for L_2 must be of the form $\mathbf{d} = (a, b, 0)$ for some scalars a and b. Since L_2 is perpendicular to L_1, direction vectors for L_2 must be perpendicular to direction vectors of L_1, and since L_1 has $(2, -3, 4)$ as a direction vector, we find that

$$\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = 2a - 3b = 0.$$

Thus $a = (3/2)b$.

Any vector solving this equation gives a direction vector of L_2, and so (to obtain integer values of a and b) we can pick $b = 2$, and thus $a = 3$. So $\mathbf{d} = (3, 2, 0)$ is a direction vector of L_2, and since L_2 passes through $(1, 3, -1)$, we obtain a parameterization of L_2 of the form

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 + 3t \\ 3 + 2t \\ -1 \end{bmatrix}.$$

The parameterizations of straight lines given above are given by vector-valued functions whose components are linear functions. But we can also parameterize curves using nonlinear functions.

Example

As we vary t over its allowed values, the vector-valued function $\mathbf{r}(t)$ traces a curve in $\mathbb{R}^n$. We will not graph functions by hand in this course. For example, consider

$$\mathbf{r}(t) = 4\cos t\mathbf{i} + 4\sin t\mathbf{j} + t\mathbf{k}, \quad 0 \leq t \leq 4\pi.$$

Then the curve it traces is illustrated below.

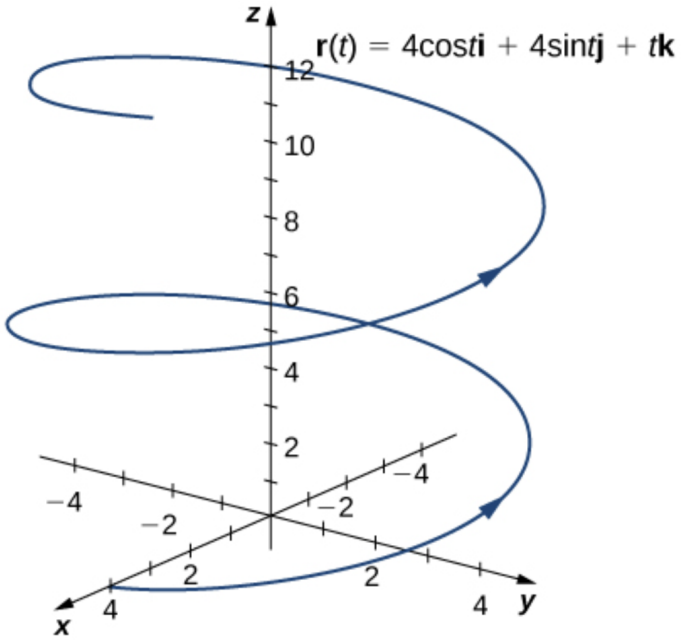


Figure. A 3D helix traced by a vector-valued function.

Activity: Evaluating a vector-valued function (U3-LO1)

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(0)$.

We substitute 0 for t in the equation above, obtaining that

$$\begin{aligned}\mathbf{r}(0) &= ((0)^2 - 3(0))\mathbf{i} + (4(0) + 1)\mathbf{j} \\ &= 0\mathbf{i} + 1\mathbf{j} \\ &= \mathbf{j}.\end{aligned}$$

We can also rewrite the equation $\mathbf{r}(0) = \mathbf{j}$ as $\mathbf{r}(0) = (0, 1)$.

Activity: Evaluating a vector-valued function (U3-LO1)

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(1)$.

We calculate that

$$\begin{aligned}\mathbf{r}(1) &= ((1)^2 - 3(1))\mathbf{i} + (4(1) + 1)\mathbf{j} \\ &= (1 - 3)\mathbf{i} + (4 + 1)\mathbf{j} \\ &= -2\mathbf{i} + 5\mathbf{j}.\end{aligned}$$

Alternatively, we can write $\mathbf{r}(1) = (-2, 5)$.

Activity: Evaluating a vector-valued function (U3-LO1)

Recall the vectors $\mathbf{i}$ and $\mathbf{j}$ in $\mathbb{R}^2$ from Standard basis. Consider the vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^3$ defined by the equation

$$\mathbf{r}(t) = (t^2 - 3t)\mathbf{i} + (4t + 1)\mathbf{j},$$

i.e., the equation

$$\mathbf{r}(t) = \begin{bmatrix} t^2 - 3t \\ 4t + 1 \end{bmatrix}.$$

Evaluate $\mathbf{r}(-4)$.

We calculate that

$$\begin{aligned}\mathbf{r}(-4) &= (16 + 12)\mathbf{i} + (-16 + 1)\mathbf{j} \\ &= 28\mathbf{i} - 15\mathbf{j}.\end{aligned}$$

So $\mathbf{r}(-4) = (28, -15)$.

We use parameterizations mainly to describe position, velocity, and tangent direction along a curve.

Limits and continuity

Definition: Limit of a Vector-Valued Function

A vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ approaches the limit $\mathbf{v} \in \mathbb{R}^n$ as t approaches a, written

$$\lim_{t \to a} \mathbf{r}(t) = \mathbf{v},$$

provided

$$\lim_{t \to a} \|\mathbf{r}(t) - \mathbf{v}\| = 0.$$

This is the limit of a scalar-valued function, and can thus be evaluated using the methods of single-variable calculus.

In practice, calculating and simplifying the expression $\|\mathbf{r}(t) - \mathbf{v}\|$ in order to evaluate the limit

$$\lim_{t \to a} \|\mathbf{r}(t) - \mathbf{v}\| = 0$$

can be quite cumbersome. Instead, the following theorem allows us to evaluate limits much more easily.

Theorem: Limit of a Vector-Valued Function

Let $\mathbf{r}(t) : \mathbb{R} \to \mathbb{R}^n$ be a function with component functions $r_1, \ldots, r_n : \mathbb{R} \to \mathbb{R}$. Then if

$$\lim_{t\to a} r_1(t) = v_1 \quad \cdots \quad \lim_{t\to a} r_n(t) = v_n,$$

and we define the vector $\mathbf{v} = (v_1, \ldots, v_n)$, then

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{v}.$$

Equivalently,

$$\lim_{t\to a} \mathbf{r}(t) = \begin{bmatrix} \lim_{t\to a} r_1(t) \\ \vdots \\ \lim_{t\to a} r_n(t) \end{bmatrix}.$$

Definition: Continuity of Vector-Valued Functions

A vector-valued function $\mathbf{r}(t)$ is continuous at point $t = a$ if

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{r}(a)$$

That is, a vector-valued function is continuous at point $t = a$ when the limit of the function as t approaches a equals the value of the function at a.

Theorem

A vector-valued function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ with component functions $r_1, \ldots, r_n$ is continuous at $t = a$ if and only if each of the component functions $r_1, \ldots, r_n$ is continuous at $t = a$.

Derivatives

Now we define the derivative of a vector-valued function from $\mathbb{R}$ to $\mathbb{R}^n$.

Definition: Derivative of a Vector-Valued Function

The derivative of a function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ is

$$\frac{d\mathbf{r}}{dt}(t) = \mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t},$$

provided the limit exists. If $\mathbf{r}'(t)$ exists, then we say $\mathbf{r}$ is differentiable at t. If $\mathbf{r}'(t)$ exists for all t in an open interval (a, b), then we say $\mathbf{r}$ is differentiable on (a, b). The vector $\mathbf{r}'(t)$ is called the tangent vector of the curve described by the function $\mathbf{r}$ at the point $\mathbf{r}(t)$.

Remark

The derivative $\mathbf{r}' : \mathbb{R} \to \mathbb{R}^n$ of a differentiable function $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ is *also* a vector-valued function.

As in single variable calculus, we will often have much more efficient methods of calculating the derivative of a function $\mathbf{r}$. But you should be able to work directly from the definition in simple examples.
Since Limit of a Vector-Valued Function tells us taking limits of functions from $\mathbb{R}$ to $\mathbb{R}^n$ is equivalent to taking the limits of each component function, taking the derivative of such a function is equivalent to taking the derivative of each component function.

Theorem

If $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$ has components $r_1, \ldots, r_n : \mathbb{R} \to \mathbb{R}$, then $\mathbf{r}$ is differentiable at t if and only if each component function $r_1, \ldots, r_n$ is differentiable at t, and

$$\mathbf{r}'(t) = \begin{bmatrix} r_1'(t) \\ \vdots \\ r_n'(t) \end{bmatrix}.$$

Example: A vector derivative from component functions (U3-LO2)
We use the theorem to calculate the derivative of the function

$$\mathbf{r}(t) = (3t^2 + 2)\mathbf{i} + (2t - 1)\mathbf{j}.$$

Both component functions are polynomials, so they are differentiable for every $t \in \mathbb{R}$. By the theorem, we differentiate each component:

$$r_1'(t) = \frac{d}{dt}(3t^2 + 2) = 6t,$$
$$r_2'(t) = \frac{d}{dt}(2t - 1) = 2.$$

Therefore,

$$\mathbf{r}'(t) = 6t\mathbf{i} + 2\mathbf{j} = \begin{bmatrix} 6t \\ 2 \end{bmatrix}.$$

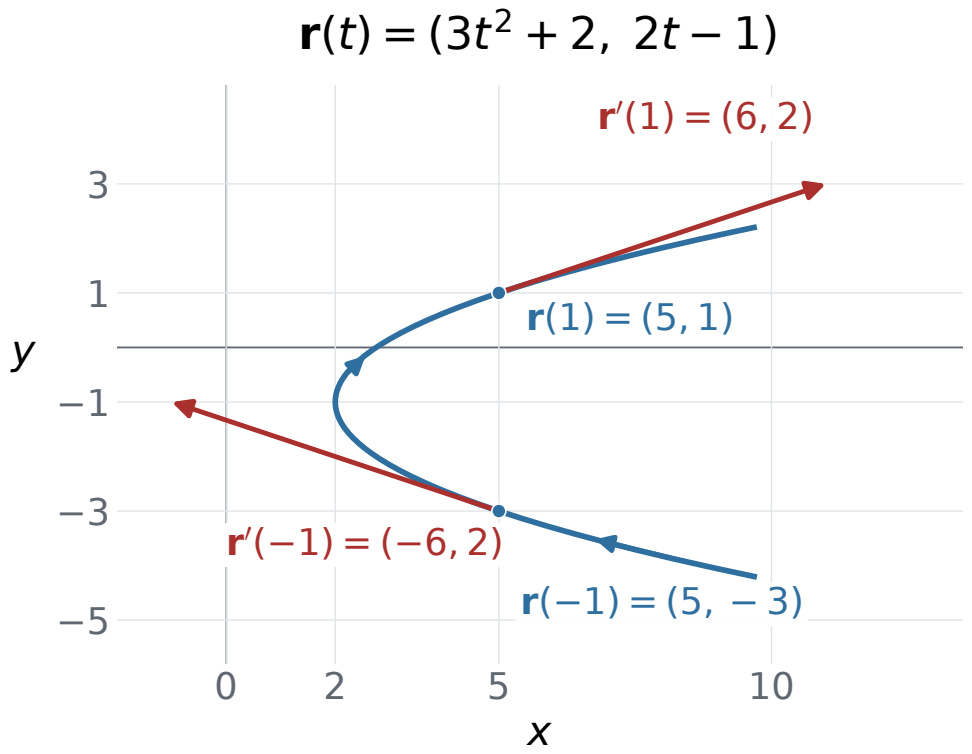


Figure. A right-opening parabola with vertex at (2, minus 1). As t increases, the curve moves upward and left along the lower branch toward the vertex, then upward and right. At r(minus 1) = (5, minus 3), a tangent vector (minus 6, 2) points upward and left to (minus 1, minus 1). At r(1) = (5, 1), a tangent vector (6, 2) points upward and right to (11, 3). Both axes use the same scale.

Because of the theorem, we can apply all the techniques from single-variable calculus to differentiate functions from $\mathbb{R}$ to $\mathbb{R}^n$.

Activity: Differentiating a vector-valued function (U3-LO2)

Find $\mathbf{r}'(t)$, where

$$\mathbf{r}(t) = e^{2t}\,\mathbf{i} + \ln(1+t)\,\mathbf{j} - (\cos t)\,\mathbf{k}$$

Using the theorem, we calculate that

$$\begin{aligned}\mathbf{r}'(t) &= (e^{2t})' \ \mathbf{i} + (\ln(1+t))' \ \mathbf{j} - (\cos t)' \ \mathbf{k} \\ &= 2e^{2t} \ \mathbf{i} + \frac{1}{1+t} \ \mathbf{j} + \sin t \ \mathbf{k}.\end{aligned}$$

If $\mathbf{r} : \mathbb{R} \to \mathbb{R}^n$, then $\mathbf{r}' : \mathbb{R} \to \mathbb{R}^n$ is also a vector-valued function of the same type, so we can consider the second derivatives $\mathbf{r}''$, i.e., the derivative of the function $\mathbf{r}'$.

Differentiation rules

Theorem

Suppose $\mathbf{u}, \mathbf{v} : \mathbb{R} \to \mathbb{R}^n$ are differentiable, c is a scalar, and $a : \mathbb{R} \to \mathbb{R}$ is a scalar-valued function. Then
Sum Rule
$(\mathbf{u} + \mathbf{v})'(t) = \mathbf{u}'(t) + \mathbf{v}'(t)$
Product Rule
$(c\mathbf{u})'(t) = c\mathbf{u}'(t)$
Product Rule
$(a\mathbf{u})'(t) = a'(t)\mathbf{u}(t) + a(t)\mathbf{u}'(t)$
Dot Product rule
$(\mathbf{u} \cdot \mathbf{v})'(t) = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

Why is this true?

We prove the dot product rule, in the two dimensional case. Let

$$\mathbf{u}(t) = u_1(t)\mathbf{i} + u_2(t)\mathbf{j}$$

and let

$$\mathbf{v}(t) = v_1(t)\mathbf{i} + v_2(t)\mathbf{j}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

Applying the sum and product rules for functions from $\mathbb{R}$ to $\mathbb{R}$, we find that

$$\begin{aligned}
(\mathbf{u} \cdot \mathbf{v})' &= (u_1 v_1)' + (u_2 v_2)' \\
&= (u_1' v_1 + u_1 v_1') + (u_2' v_2 + u_2 v_2') \\
&= (u_1' v_1 + u_2' v_2) + (u_1 v_1' + u_2 v_2') \\
&= \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'.
\end{aligned}$$